Observational Indistinguishability and Integrity Blind Regions Across the Evidence Boundaries of Hybrid Quantum-Classical Kernel Workflows

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Abstract—Integrity auditing asks whether the evidence exposed at a workflow boundary can distinguish a consequential intervention from the baseline. We formalize this question for hybrid quantum-classical workflows as observational indistinguishability under an explicit information set and explicit trusted references. A structural blind region is the set of interventions whose observable view equals the baseline view; it is monotone under refinement of the view and is closed exactly by added evidence that separates the intervention class. The evaluation-label path is a protected boundary: for a fixed predictor, label-only changes leave feature and prediction evidence invariant, prior-preserving changes also leave label marginals invariant, and any change of the reported balanced accuracy changes the confusion matrix, so a material label corruption is always separable in the joint item-label view yet detectable at will only against an item-aligned trusted reference. Across CICIDS2017, UNSW-NB15 and ToN-IoT in eight fixed environments (11,400 deduplicated observations, 3,600 label interventions) the blind regions hold without exception and survive null calibration. A decision-level calibration lowers the false-alarm rate of batch-level regimes from 0.125–0.259 to 0.061–0.079, and an end-to-end `allow/hold/block` evaluation over 24,000 frozen observations shows that batch-level regimes with feature evidence serve 4,322–4,494 of 7,008 materially changed results, the label-marginal regime serves all of them, and the trusted item-aligned regime serves none. A 165-cell simulator gate maps the same lattice onto circuit, kernel and estimation evidence, and a four-contract prototype executes the composed fail-closed decision.

Index Terms—hybrid quantum-classical systems, integrity auditing, observability, quantum kernels, runtime assurance

1 Introduction

Hybrid quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries. Classical records are acquired, cleaned, projected and encoded; a circuit and an execution stack produce quantum evidence; a classical learner turns a kernel into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label or reporting path, and security analyses of hybrid systems call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

Robustness benchmarking collapses this chain into a performance change. That endpoint cannot say what changed or whether the available evidence could have revealed the change. Consider the evaluation boundary: a deployed predictor receives the same feature batch and emits the same predictions, but the labels used to score the batch are corrupted. Balanced accuracy moves although the model has not. Feature and prediction monitors cannot observe the event because their inputs are identical; label counts expose an unbalanced corruption, but a balanced exchange of class identities preserves them. Calling the event stealthy omits the decisive qualifier: stealthy to which auditor, holding which evidence, trusting which reference?

That qualifier has a long history outside machine learning. Control theory characterizes undetectable and stealthy attacks as inputs that are indistinguishable from nominal behaviour through the monitor's outputs [3]–[5], quantifies the detectability–impact trade-off [6], [7], and treats observability as a property of the monitored system rather than of the attack [8]. This article transports that discipline to the evidence chain of a hybrid learning workflow and makes three moves that the control-theoretic literature does not: it treats the *evaluation-label path* as a protected boundary with its own blind regions; it separates structural indistinguishability (the view is identical) from statistical undetectability (the view differs but not beyond the null variability of fresh batches) and shows that only an item-aligned *trusted reference* converts the second into an exact test; and it composes the resulting coverage into an executable, calibrated, fail-closed decision.

The contribution is one package:

- an observation model of hybrid workflows in which auditability is a property of (intervention class, information set, trusted references), with monotonicity, closure and materiality results whose earlier label-path propositions are corollaries, and minimal counterexamples that separate integrity violation, marginal invariance, confusion-matrix invariance and conclusion change;
- an explicit adversary and failure model that assigns

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every executed intervention to fault robustness, integrity corruption, adversarial attack or adaptive attacker, with the roots it cannot write;
- multi-dataset, fixed-OOD validation with verified deduplication, cluster-level uncertainty and preregistered null calibration in which the exact blind regions survive;
- a decision-level (family-wise) calibration that replaces per-sensor budgets by a false-alarm budget of the decision, and an end-to-end `allow/hold/block` evaluation on 24,000 frozen observations whose primary endpoint is the number of materially changed results a policy would serve; and
- a bounded quantum-specific layer and a four-contract prototype that execute the composed decision within a local research boundary.

No constituent is claimed to be individually sufficient novelty. The propositions are finite-batch statements; contracts, hash chains and semantics-preserving tests have clear prior art; and the secondary quantum-versus-classical impact profile is neither a quantum-advantage result nor a vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference and operational Fleet Management are outside the claim.

2 Related Work and Positioning

2.1 Stealth, Detectability and Observability in Secure Systems

The idea that detectability depends on what the monitor can see is established. Pasqualetti *et al.* characterize undetectable and unidentifiable attacks on cyber-physical systems through the observability of the monitored dynamics [3]; Teixeira *et al.* organize adversaries by model knowledge, disclosure and disruption resources and derive zero-dynamics and replay attacks that are stealthy to residual detectors [4]; Liu *et al.* exhibit false-data injections that bypass bad-data detection because they lie in the estimator's range space [5]. Later work quantifies how much an ε-stealthy attacker can degrade performance and how to limit that impact [6], [7], and surveys physics-based detection as a question of which residuals a monitor can form [8]. Runtime assurance routes a decision through a verified simple controller when a monitor flags the complex one [9], runtime verification checks executions against specifications [10], and integrity checkers compare stored digests against trusted baselines [11].

This article does not claim the general insight that detectability is information-relative. It differs in the object and in the missing pieces: the monitored system is an evidence chain whose protected boundaries include the ground truth used to score the model; the blind regions are derived from views rather than from linear dynamics; the role of a trusted reference is made explicit as the difference between statistical and exact separation; and the coverage is composed into a calibrated decision and evaluated end to end.

2.2 Monitoring, Labels and Evaluation Integrity

Label shift estimation targets a changed class prior under source-target assumptions [12]; MLDemon and uncertainty-aware monitoring combine unlabeled observations with selectively acquired labels [13], [14]; an evidence-sufficiency framework maps proxy coverage under delayed ground truth and formalizes the inability of feature proxies to expose concept drift [15]. Northcutt *et al.* document pervasive label errors in test sets and their effect on benchmark rankings [16]; Arp *et al.* catalogue sampling, labeling, metric and threat-model pitfalls in security benchmarks [17]; Bajaj names *evaluation blindness*, a measurement indistinguishable from a healthy state while the system fails, and documents its prevalence in production incidents [18]. These works treat labels as available, noisy or late. Here the label path is an asset under intervention, availability is separated from trust, and the evidence needed to identify a corruption is derived, not assumed.

2.3 Hybrid and Quantum Software Integrity

Volya *et al.* decompose classical-quantum systems into control, compilation, execution and measurement surfaces [1]; surveys extend the threat space to malicious compilation, hardware faults, cloud exposure and side channels [2], [19]; Quantum Vulnerability Factor scores circuit-output sensitivity to faults [20] and Quantum Leak demonstrates timing leakage in cloud services [21]. QCIVET combines stage contracts, hash-chained traces, behavioural subtyping and calibrated observable-deviation tests with QPU validation [22]; QML-PipeGuard fingerprints the quantum stage through expectation values under a tomographically structured measurement family, absorbs benign calibration drift within a calibrated tolerance and detects channel substitution on an IBM Heron processor [23]; a multi-level framework evaluates structural, interaction and behavioural circuit-integrity metrics [24]; and a cross-provider analysis proposes a unified provenance contract for quantum jobs [25]. Quantum software testing addresses the oracle problem through differential, metamorphic and equivalence-modulo-input relations [26]–[28], and compiler verification, program logics and runtime assertions supply formal and runtime mechanisms [29]–[33]. We make no priority claim for contracts, hash chains, stage decomposition, semantic probes or fail-closed verification. Their unit is the quantum stage or the pipeline trace; ours is the information set that makes a changed asset distinguishable, with the evaluation-label path as a first-class boundary, coverage mapped across data environments, and the contract executing a calibrated map.

Table 1 states the combination claim. Prior work establishes information-relative detectability for controlled dynamics, proxy monitoring for late labels, and contract-based integrity for quantum stages. This work differs in deriving blind regions from views of a learning workflow whose protected boundaries include the evaluation labels, in separating structural from statistical blindness through explicit trusted references, and in composing the coverage into a decision whose false-alarm budget and unsafe-allow count are measured.

2.4 Relation to Companion Work

Three companion studies by the authors share infrastructure or vocabulary and are disclosed to delimit the claim. *Sharp target-domain certificates* [34] bounds the advantage of a fixed quantum-kernel candidate under distribution shift

TABLE 1
Positioning against the closest lines of work. A dash means that the property is not the work's declared object, not that the work is deficient.

Line of work	Evidence conditioning	Evaluation-label boundary	Trusted reference	Validation and response
Stealthy attacks on CPS [3]–[6]	Observability of linear dynamics	–	Model and sensors trusted	Residual detectors; impact bounds
Runtime assurance and integrity monitoring [9]–[11]	Specification or baseline digest	–	Verified controller or stored baseline	Switching, monitors, digests
Hybrid threat taxonomies [1], [2]	–	–	–	Taxonomy; no executable response
QCIVET [22]	Stage contracts/observables	–	Reference observables	Calibrated noise and QPU; verification engine
QML-PipeGuard [23]	Calibrated observable tolerance	–	Fingerprint of the channel	Two-qubit QSVM on IBM Heron
Evidence sufficiency [15]; evaluation blindness [18]	Proxy coverage; detectability taxonomy	Delayed or silent outcomes	–	One dataset; incident corpus
This work	**Views, refinement order, structural versus calibrated blindness**	**First-class protected boundary**	**Explicit; exact versus statistical separation**	**Three datasets, eight environments, calibrated fail-closed policy evaluated end to end**

with partial target labels; it shares data sources, fidelity kernels and a fail-closed prediction lock, but its object is an identification interval, not the observability of interventions. *Conditional validity of quantum event classifiers* [35] is the closest conceptual relative: it conditions fail-closed verdicts on a declared information set and exhibits exact sensor blindness to weight-only nuisances in collider data. The present article applies the conditioning to the integrity of the evidence chain under controlled interventions, with trusted references, calibrated policy and executable response. *Candidate comparability before promotion* [36] decides whether a challenger should replace a deployed detector after a drift alarm; it shares the benchmarks and label-free monitors, but its unit is a sequential promotion decision. None of the propositions, interventions, tables or experiments reported here appears in those works.

3 Observation Model and Blind Regions

The complete statement with proofs is in the supplement; this section states the objects and the results the evidence tests. The results are elementary by design: their role is to make exact which evidence separates which intervention class, so that the gates test statements rather than intuitions.

3.1 States, Interventions and Materiality

A workflow state is the tuple of artifacts of one evaluation run,

$$s = (X, P, C, K, E, f, y, R) \in \mathcal{S}, \tag{1}$$

with acquired features X, preprocessing P, circuit or feature map C, estimated kernel K, execution metadata E, fitted predictor f, evaluation labels y and reported result R; $\tilde{X} = P(X)$ is the evaluation representation and $\hat{y} = f(\tilde{X})$ the predictions, indexed by item identity $i = 1, \ldots, n$. An intervention is a map $a : \mathcal{S} \to \mathcal{S}$; the baseline is s_0 and the intervened state $s_a = a(s_0)$. Classes used below: label-only T_y (changes y item-wise, fixes everything else), its prior-preserving subclass T_y^π (also preserves the class histogram), and feature-side mechanisms that change $\tilde{X}$ and, through f, possibly $\hat{y}$ and R. The conclusion is $R(s)$, balanced accuracy of $\hat{y}$ against y, and its signed change is

$$\Delta_R(a) = R(s_0) - R(s_a). \tag{2}$$

An intervention is *material at level* τ if $|\Delta_R(a)| \geq \tau$; $\tau \to 0^+$ means any change of the reported conclusion. Earlier versions of this evidence used the positive part $\max(\Delta_R, 0)$; the signed endpoint is used here because an apparent improvement caused by an integrity failure is a conclusion change.

3.2 Views, Refinement and Trusted References

An information set $\mathcal{I}$ is a view map $V_\mathcal{I} : \mathcal{S} \to \mathcal{O}_\mathcal{I}$; the auditor observes $V_\mathcal{I}(s)$ and nothing else. The regimes studied are $\mathcal{I}_X$ (the multiset of rows of $\tilde{X}$), $\mathcal{I}_{XF}$ (the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$), $\mathcal{I}_{Y_m}$ (the class-count vector of y), $\mathcal{I}_{XFY}$ (the multiset of triples $(\tilde{x}_i, \hat{y}_i, y_i)$) and $\mathcal{I}_Q$ (circuit, kernel and execution evidence). $\mathcal{I} \sqsubseteq \mathcal{I}'$ (refinement) iff $V_\mathcal{I} = \pi \circ V_{\mathcal{I}'}$ for some π; hence $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF} \sqsubseteq \mathcal{I}_{XFY}$ and $\mathcal{I}_{Y_m} \sqsubseteq \mathcal{I}_{XFY}$, while $\mathcal{I}_{Y_m}$ is incomparable with $\mathcal{I}_X$ and $\mathcal{I}_Q$ is a separate branch. The join $\mathcal{I} \vee \mathcal{W}$ is the pair of views.

A *reference* is a stored value $\rho = V_\mathcal{J}(s_0)$ of a view of the baseline. It is *trusted* under the assumption that no intervention in the class under study can alter it, and *item-aligned* when $V_\mathcal{J}$ is indexed by item identity rather than a multiset, so that it supports item-wise comparison. $\mathcal{I}^\star$ denotes a regime augmented with a trusted item-aligned reference of its own baseline view. A label being available in $\mathcal{I}_{XFY}$ says nothing about whether it is trusted: if the label store is the asset under attack, the auditor sees y_a, not y_0.

3.3 Equivalence, Sensors and Blind Regions

Definition 1. $s \sim_\mathcal{I} s'$ iff $V_\mathcal{I}(s) = V_\mathcal{I}(s')$. A sensor admissible in $\mathcal{I}$ is a measurable $S : \mathcal{O}_\mathcal{I} \to \mathbb{R}^k$, written $S(s) = S(V_\mathcal{I}(s))$.

Lemma 1 (structural blindness). If $s_a \sim_\mathcal{I} s_0$ then $S(s_a) = S(s_0)$ for every sensor admissible in $\mathcal{I}$ (path-wise for a seeded randomized sensor; in distribution for a fresh seed).

Definition 2. For a class $\mathcal{A}$, the structural blind region of $\mathcal{I}$ is $B_\mathcal{I}(\mathcal{A}) = \{a \in \mathcal{A} : a(s_0) \sim_\mathcal{I} s_0\}$ and the sensor blind region of a family $\mathcal{S}$ is $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) =$

$S(s_0)\ \forall S \in \mathcal{S}\}$. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ separates the orbit of $\mathcal{A}$; sensor blindness thus decomposes into information-level blindness and sensor insufficiency. The structural and sensor auditability gaps at level τ are

$$G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau,\ a \in B_{\mathcal{I}}\}, \quad G_{\mathcal{I},\mathcal{S}}(\tau) \supseteq G_{\mathcal{I}}(\tau). \tag{3}$$

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$ for every τ.

Proof: $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ implies $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. □

Corollary 1 (label-path boundaries). Let f be fixed and deterministic. (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$: every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change. (b) $T_y^{\pi} \subseteq B_{\mathcal{I}_{Y_m}}$: every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change. These are the two propositions of the earlier version of this evidence, now instances of Lemma 1 and Proposition 1.

Proposition 2 (closure). $B_{\mathcal{I}\vee\mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$. A class $\mathcal{A} \subseteq B_{\mathcal{I}}$ becomes completely separable after adding $\mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$: the added evidence must be injective on the orbit of the class. Consequently no view factoring through $(\tilde{X}, f(\tilde{X}), \mathrm{hist}(y))$ separates any $a \in T_y^{\pi}$; the multiset $\mathcal{I}_{XFY}$ separates it unless labels are permuted only among items with identical $(\tilde{x}, \hat{y})$, and $\mathcal{I}^{\star}_{XFY}$ separates every non-identity relabeling.

Proposition 3 (materiality forces separability). Let $R = g(M)$ be a function of the confusion matrix $M(s)$, as balanced accuracy is. If $a \in T_y$ and $\Delta_R(a) \neq 0$ then $M(s_a) \neq M(s_0)$; hence $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for every $\tau > 0$.

Proof: Contrapositive of $M(s_a) = M(s_0) \Rightarrow R(s_a) = R(s_0)$; the confusion view is a coarsening of the multiset of triples. □

3.4 Three Auditor Classes

A *reference-anchored* auditor holds a trusted item-aligned reference ρ and computes $D(s) = d(V_{\mathcal{J}}(s), \rho)$; then $D(s_0) = 0$ exactly, the rule "fire iff $D > 0$" has zero false-alarm probability, and separability is sufficient for detection. An *anchor-free* (batch-level) auditor holds only a null model of $V_{\mathcal{I}}$ over fresh clean batches and decides with a calibrated test at budget α; for a separable a its detection probability is the power against the shift relative to batch-to-batch variability, and separability is not sufficient. A *post-hoc certifier* recomputes a view from inputs it trusts (the evaluation contract recomputes R from $(\hat{y}, y)$) and needs trusted inputs rather than a stored value.

Proposition 4. (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$, $a \in B_{\mathcal{I}}$ implies $\mathrm{fire}(s_a) = \mathrm{fire}(s_0)$: the calibrated detection rate equals the false-alarm rate. (ii) A reference-anchored auditor detects every $a \notin B_{\mathcal{J}}$ with probability one.

Proposition 5 (union versus family calibration). If sensors $S_1, \ldots, S_m$ are each calibrated so that $\Pr_0[S_j > t_j] \leq \alpha$, the rule "fire iff some $S_j > t_j$" has null firing probability in $[\alpha, \min(1, m\alpha)]$. Let $U = \max_j F_j(S_j)$ with F_j the calibration distribution of S_j and q the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration value of U. If the calibration draws and the audited batch are exchangeable, $\Pr_0[U > q] \leq \alpha$ [37]–[39].

TABLE 2
Minimal counterexamples and their witnesses in the frozen expansion (count / denominator). $\hat{y}$ predictions, y labels, M confusion matrix, R balanced accuracy.

ID	Construction and what it separates	Witnesses
C1	Swap y_i, y_j with $\hat{y}_i = \hat{y}_j$: M, hist, R invariant; item identity changes	808 / 3,600
C2	Relabeling that changes M: $\hat{y}$ invariant, R changes	2,617 / 3,600
C3	Swap y_i, y_j with $\hat{y}_i \neq \hat{y}_j$: hist invariant, M and R change	1,059 / 1,800
C4	Compensating flips: M changes, R invariant	175 / 3,600
C5	Feature change that flips predictions but not R	296 / 7,200
C6	Feature change that crosses no decision boundary	2,513 / 7,200
C7	Intervention that raises R (apparent improvement)	1,534 / 10,800
C8	Material label change always changes M (Prop. 3)	2,617 / 2,617

Proposition 6 (no local guarantee without an uncontrolled root). If the intervention class rewrites the reference consistently with the asset, $a : (s, \rho) \mapsto (a(s), V_{\mathcal{J}}(a(s)))$, the augmented auditor is a function of the state alone and Lemma 1 applies to $\mathcal{I} \vee \mathcal{J}$: for $\mathcal{J}$ factoring through $(\tilde{X}, f(\tilde{X}), y)$ every $a \in T_y^{\pi}$ is in the joint blind region. Local identification of a label-path intervention requires an item-aligned root the class cannot write.

3.5 Counterexamples

Table 2 lists minimal constructions that separate the events a robustness number conflates, with the number of frozen expansion observations that realise each of them. C1 is an integrity violation with no conclusion impact, visible only item-wise against a trusted reference; C2–C4 separate marginal invariance, confusion-matrix invariance and conclusion invariance; C5–C6 do the same on the feature side; C7 shows that the clipped "harm" endpoint of the earlier evidence hid 1,534 conclusion changes, 433 of them on the label path; C8 is the empirical face of Proposition 3.

3.6 The Quantum Branch

The quantum gate of Section V-C uses the same lattice. Views of $\mathcal{I}_Q$ are the provenance hash $h(C)$, the kernel K, its algebraic invariants $\mathrm{alg}(K)$ (a coarsening of K), the repeated-estimation discrepancy, and the outputs $f_K(\tilde{X})$ (a coarsening of K through the decision). Approved transpilation is in B_K but not in B_h: separability in the provenance view is not harm, so the policy needs an approved equivalence class implemented as semantic equality of probe kernels against a trusted reference. A positive-semidefinite (PSD)-preserving substitution is in B_{alg} but not in B_K: sensor-level blindness closed by a reference-anchored comparison. Output monitoring is blind to every change that crosses no decision boundary. Hash chaining of the audit envelope is a reference-anchored comparison over the record itself; it is tamper-evident but its root is not assumed uncontrolled, so Proposition 6 applies.

4 Adversary and Failure Model

Table 3 assigns every executed intervention to one of four classes and states the roots it cannot write and the regime that identifies it; the complete per-class record (actor, capability, access point, knowledge, objective, budget, alterable assets,

TABLE 3
Adversary and failure model of the executed interventions. Roots: assets the class cannot write. Identifying regime: the least evidence that separates the class in the frozen design; "exact" means an item-aligned trusted reference.

Mechanism (boundary)	Class	Capability	Roots it cannot write	Identifying regime	Supported claim
Random label flip (evaluation/report)	Integrity corruption	rewrite r of y at random	X, P, C, K, f; item-aligned $y^\star$ if declared	exact only; $\mathcal{I}_{Y_m}/\mathcal{I}_{XFY}$ batch 0.00–0.03	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}$ (Cor. 1a); not detected at budget by marginal or batch joint evidence at these rates
Prior-preserving label flip (evaluation/report)	Adaptive attacker	pairwise swaps preserving class counts	as above	exact only; $\mathcal{I}_{XFY}$ batch 0.005–0.012	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}$ (Cor. 1); every material instance changes M (Prop. 3)
Confusion-preserving relabeling (discussed)	Adaptive attacker	swaps among equal predictions	as above	exact only	integrity violation without conclusion impact (C1)
Feature sign flip (acquisition/preprocessing)	Integrity corruption	negate p of $\tilde{X}$ entries	training data, fitted model, reference features	$\mathcal{I}_X$ 0.09–0.72; $\mathcal{I}_{XF}$ 0.15–0.85; exact 0.79–0.91	partially separable; residual is strength-dependent
Mean shift, scaling drift (acquisition/preprocessing)	Fault robustness	shift or scale all features	as above	$\mathcal{I}_X$ 1.00 by cluster fingerprint	fully detected in this design by a fragile fingerprint
Feature dropout with imputation (preprocessing)	Fault robustness	impute p of entries	as above	batch 0.00–0.04; exact 0.49–0.76	separable in $\mathcal{I}_{XF}$ yet statistically undetectable without a reference
Cluster-preserving perturbation (discussed)	Adaptive attacker	perturb outside clusters	as above	exact	predicted to evade $\mathcal{I}_X$ at batch level; not measured
Circuit-parameter mutation; repetition change (design/transpilation)	Adversarial attack	alter C before execution	reference circuit hash, probe kernel K_0	provenance and semantic 15/15	identified by reference-anchored provenance and semantics; outputs miss sub-decision changes
Benign transpilation; common-unitary rewrite	Approved equivalence	compiler pass	–	hash changes 15/15, K unchanged	separability in $h(C)$ is not harm
Asymmetric, diagonal-eroded kernel edit (post-processing)	Integrity corruption	edit K after estimation	reference kernel, algebraic invariants	algebra 15/15	detected by structural invariants without a reference
PSD-preserving kernel substitution (post-processing)	Adaptive attacker	valid-looking K	reference kernel, circuit hash	algebra 0/15; provenance and semantic 15/15	sensor-level blindness closed by reference anchoring
Binomial shot emulation (execution)	Fault robustness	finite-shot estimate	–	repeated estimation 15/15	held for adjudication; not hardware evidence
Post-hoc envelope edit (record)	Adversarial attack	rewrite stored contracts	none assumed	hash chain	tamper-evident only; no identity or non-repudiation

observable set, supported claim) is in the artifact. Evaluation-label classes stand for corruption of the ground-truth store or of the label join; feature-side classes for corruption or drift of the acquisition path feeding both branches; circuit and kernel classes for corruption of the circuit/parameter store, the compiler output or the post-estimation kernel; shot emulation for legitimate estimation uncertainty. Equal weighting is a stress profile, not a threat distribution, and no prevalence is estimated.

The evidence trusts the local runtime and host, the training data and fitted model, the reference input and preprocessing declaration, the reference circuit and probe kernels, SHA-256 collision resistance, the item-level label reference only where $\mathcal{I}^\star_{XFY}$ is declared, and the code that builds and verifies the envelope. Compromise of the operating system, runtime or verifier; provider identity, forged job or calibration metadata and real provider attestation; malicious schedulers, physical mapping and unapproved passes on hardware; device-calibrated noise, crosstalk, multi-tenant interference and QPU faults; side channels and circuit confidentiality; denial of service; and the operational Fleet Management System are outside this study.

5 Methodology

5.1 Data, Pipelines and Interventions

The frozen first gate uses a balanced binary subset of CICIDS2017 [40]: 3,000 rows, 77 numeric columns and 128/128 capped training/evaluation samples. The expansion adds a 256/256 CICIDS scale gate; balanced ID subsets of UNSW-NB15 [41] and ToN-IoT [42]; four temporal CICIDS source-target pairs; and one UNSW temporal pair, all at 128/128 and projected dimensions 8, 10 and 12 (eight fixed environments E1–E8, supplement). They are not a probability sample of deployments. Train-defined preprocessing is applied without evaluation labels: TruncatedSVD per branch, training-fitted standardization for the RBF support-vector classifier (SVC) and frozen angular scaling to $[0, 2\pi]$ for the fidelity-kernel branches (ZZ, PauliXYZ and a Z/PauliXZ-equivalent profile, one repetition). Gate 1 uses the Qiskit Machine Learning reference evaluator [43]; the expansion uses a tagged exact-statevector evaluator validated against it at 10^{-10} on unit matrices and by an end-to-end replay (supplement). Neither is shot-noise or QPU evidence.

The intervention suite has six mechanisms at strengths 0.02, 0.05 and 0.10: feature sign flip, feature-wise mean shift, scaling drift, feature dropout with median imputation, random evaluation-label flip and prior-preserving evaluation-label flip. Feature-only sensors are feature Jensen–Shannon divergence (JSD), maximum mean discrepancy (MMD) and the Kolmogorov–Smirnov (KS) rejection rate; prediction-aware evidence adds score JSD, prediction-rate shift and prediction JSD; label-marginal evidence uses prior shift and label JSD; joint-outcome evidence uses two confusion-profile statistics. Item-aligned sensors (prediction disagreement, confusion-profile deltas, item-level label mismatch) exist only against a reference of the same items and have an exact-zero null.

5.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model seeds; its 7,980 raw rows contain 3,420 exact repeated SVC rows, leaving 4,560 unique observations. The expansion crosses the same five split seeds with two model seeds; all 360 job pairs completed and 13,680 raw rows contain 2,280 verified repeats, leaving 11,400 unique observations. Coverage endpoints are exact counts in the prespecified design. For the secondary model-profile endpoint the five split seeds are the inferential units with two-sided 95% Student-t intervals (four degrees of freedom), describing within-environment resampling only.

5.3 Null Calibration and Decision-Level Calibration

Gate N (preregistered in artifact 1.1.0) calibrates the ten distributional sensors per (environment, dimension, model) cell at a nominal per-sensor $\alpha = 0.05$: the threshold is the 191st of 200 clean calibration draws taken from one half of the held-out evaluation pool, and false alarms are measured on 200 draws from the disjoint other half. Gate F (preregistered in artifact 1.2.0, before any analysis) calibrates each regime as a family: the family score of a batch is the maximum over the regime's sensors of the fraction of calibration values strictly below the observed value, and the regime fires when the score exceeds the 191st smallest calibration family score (Proposition 5). Thresholds use calibration draws only. Because the 20 draws of a run are overlapping subsets of one pool half, the inferential unit for false-alarm rates is the (environment, split-seed) cluster: pooled rates over draws are descriptive, and intervals are five-cluster t intervals per environment. The earlier binomial intervals are descriptive for the same reason.

5.4 Decision Layer and End-to-End Evaluation

Gate D composes sensors, calibrated thresholds, the information regime, the trusted-reference status, a materiality threshold and the declared residual blind region into one `allow/hold/block` decision. `block` is reserved for violations of an exact invariant against a trusted reference; `hold` answers statistical evidence or missing mandatory evidence; the decision composes with the four frozen contracts by the maximum in `allow` < `hold` < `block`. Four policies are fixed: P0 serve-always; P1 the uncalibrated union rule of Gate N; P2 the family-calibrated rule; P3 a strict fail-closed variant that holds whenever a protected boundary (feature, prediction, label) is neither exactly nor statistically covered by the regime. Five regimes are evaluated: $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}, \mathcal{I}_{XFY}$ (batch-level) and $\mathcal{I}^{\star}_{XFY}$ (item-aligned trusted reference). The observation set is frozen: 12,000 clean evaluation draws, 1,200 near-null in-place controls, and the 10,800 intervened observations of the eight environments, 7,008 of which

TABLE 4
Verified coverage counts after computational-unit deduplication. $\mathcal{I}_{Y_m}$ applies to prior-preserving rows only; the signed conclusion change is recomputed from the frozen tables by the artifact verifier.

Verified quantity	Gate 1	Expansion
Unique observations	4,560	11,400
Label-only / $\mathcal{I}_{XF}$ blind	1,440 / 1,440	3,600 / 3,600
Prior-preserving / $\mathcal{I}_{Y_m}$ blind	720 / 720	1,800 / 1,800
Lowered / unchanged / raised R	1,276/105/59	2,184/983/433
Changed R / joint response	1,335 / 1,335	2,617 / 2,617

change the reported balanced accuracy (primary materiality; $\tau = 0.02$ and 0.05 as sensitivity). The primary endpoint is the unsafe allow: a material observation decided `allow`. Ten consistency checks fail closed, and every table and figure is generated from the manifested outputs.

5.5 Quantum Gate and Executable Prototype

A separate simulator gate isolates the quantum boundary with CICIDS, 64/64 samples, five split seeds, dimensions 4, 6 and 8 and 11 conditions (165 cells): clean, benign level-1 transpilation and a common post-feature-map rewrite as controls; data-dependent RZ mutations at 0.02 and 0.10; a feature-map repetition change; asymmetric, diagonal-eroded and PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots. Each cell records canonical OpenQASM 3 provenance, kernel hashes, semantic differences against a trusted reference, algebraic checks, repeated-estimation discrepancy and prediction changes. The prototype instantiates four ordered contracts (input/preprocessing, circuit/kernel, execution/result, evaluation/report) in an SHA-256 chain over one compact CICIDS cell and six prespecified scenarios; it evaluates composition and decision logic, not throughput.

6 RESULTS

6.1 Exact Label-Path Blind Regions

All 3,600 expansion and 1,440 Gate-1 evaluation-label observations have zero prediction disagreement and exactly unchanged feature JSD, MMD, KS rejection and score JSD; all 1,800 and 720 prior-preserving observations also have zero prior shift and label JSD (Table 4). The signed conclusion change is not one-sided: in the expansion 2,184 label observations lower the reported balanced accuracy, 983 leave it unchanged and 433 raise it (Gate 1: 1,276, 105, 59). Every one of the 2,617 label observations with a changed conclusion has non-zero item-aligned confusion evidence, and every one of the 2,184 with a lowered conclusion, the count reported by the earlier version, is among them. The boundary reproduces without exception across the eight environments and every model profile: the intervention has not become less stealthy; the auditor has acquired a more informative view.

6.2 Calibrated Coverage and Decision-Level Calibration

Figure 1(a) applies the frozen per-sensor thresholds of Gate N to the frozen interventions. Over the 12,000 evaluation draws the per-sensor false-alarm rates lie between 0.029 and 0.069, but the batch-level regime rule "fire if any sensor fires" is not calibrated at the level of the decision: its rates are 0.125, 0.203, 0.048 and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$, which join three, six, two (perfectly dependent) and ten sensors. The family-calibrated rule of Gate F brings them to 0.061, 0.073, 0.048 and 0.079 (Fig. 2(a)); the per-environment cluster means range from 0.046 to 0.146 for $\mathcal{I}_{XFY}$, with the excess over the nominal 0.05 concentrated in the environments whose pool halves are smallest or most duplicated (E1 at 0.146, ToN-IoT). The residual excess measures the violation of the exchangeability premise by overlapping draws and by the calibration/evaluation halves being different rows of one staged table; it is reported as executed. The expectation recorded before analysis, "near 0.05 in every regime", was met only approximately.

The exact blind regions are not threshold artifacts and not calibration artifacts: on all 3,600 label observations the calibrated and the family-calibrated $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules, and on all 1,800 prior-preserving observations the $\mathcal{I}_{Y_m}$ rules, fire zero times (Proposition 4(i)). The batch-level $\mathcal{I}_{XFY}$ rules detect 43 (union) and 16 (family) of the 2,617 material label observations, 0.016 and 0.006, although every one of them is separable (Proposition 3, C8): a 2–10% relabeling of a 128-row batch shifts the confusion profile by less than its variability across fresh clean batches. The item-aligned auditor detects all 2,617 (Proposition 4(ii)). The contrast measures the informational value of item-level provenance; it is not an oracle, because it costs a trusted item-aligned copy of the baseline view. Feature-side coverage is mechanism-dependent: mean shift and scaling drift are detected in every cell because the standardized projected features hold tight clusters (in every environment some feature keeps 69–87% of rows within 0.01 standard deviations) that an in-place perturbation smears and a fresh batch reproduces, a legitimate but fragile fingerprint that also fires on the near-null controls (43–68% with the family rule); feature dropout with imputation is nearly invisible at the batch level (0–4%) yet changes predictions in 49–76% of cells. The family rule costs little detection: containment of material observations falls from 0.40 to 0.38 in $\mathcal{I}_{XF}$ at the primary threshold and from 0.70 to 0.67 at $\tau = 0.05$.

6.3 End-to-End Decisions

Table 5 and Fig. 2(b) report the end-to-end evaluation. Under P0 every one of the 7,008 material observations is served. Under the batch-level regimes the calibrated policies serve 4,494 ($\mathcal{I}_X$), 4,327 ($\mathcal{I}_{XF}$) and 4,322 ($\mathcal{I}_{XFY}$) of them with decision false-alarm rates 0.061, 0.073 and 0.079, whereas the uncalibrated union serves 4,429, 4,231 and 4,180 at 0.125, 0.203 and 0.259: the union rule buys 0.40 versus 0.38 containment in $\mathcal{I}_{XFY}$ at more than three times the false-hold cost. The unsafe allows are dominated by structure, not by thresholds: all 2,617 material label observations are served under $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ (residual blind region), 2,601 of them under batch-level $\mathcal{I}_{XFY}$ (statistically undetectable), and 1,053 of 1,079 material dropout observations under $\mathcal{I}_{XF}$; $\mathcal{I}_{Y_m}$ contains nothing in this suite. The trusted item-aligned regime serves none of the 7,008 material observations with zero false holds, blocks every non-identity relabeling and every prediction change, and holds or blocks 0.55 of the near-null controls through its batch-level component. The strict policy P3 makes information-set conditionality literal: in $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and $\mathcal{I}_{Y_m}$ the label boundary is unverified, so nothing is served; in $\mathcal{I}_{XFY}$ and $\mathcal{I}^{\star}_{XFY}$ it coincides with P2. The choice between P2 and P3 is therefore a declared risk decision, not a detector property. The composition with the six frozen contract envelopes reproduces their actions (supplement).

TABLE 5
End-to-end decisions on the frozen observations (primary materiality $|\Delta_R| > 0$; 7,008 material of 10,800 intervened observations; decision false-alarm rate on 12,000 clean draws, or on the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$). P0 serve-always allows all 7,008 material observations in every regime. Residual blind: material observations structurally indistinguishable from the reference under the regime.

Regime	Policy	Dec. FPR	Unsafe allow	Contain.	Res. blind
$\mathcal{I}_X$	P1 union	0.125	4,429	0.37	2,617
$\mathcal{I}_X$	P2 family	0.061	4,494	0.36	2,617
$\mathcal{I}_X$	P3 strict	1.000	0	1.00	2,617
$\mathcal{I}_{XF}$	P1 union	0.203	4,231	0.40	2,617
$\mathcal{I}_{XF}$	P2 family	0.073	4,327	0.38	2,617
$\mathcal{I}_{XF}$	P3 strict	1.000	0	1.00	2,617
$\mathcal{I}_{Y_m}$	P1 union	0.048	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P2 family	0.048	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P3 strict	1.000	0	1.00	5,450
$\mathcal{I}_{XFY}$	P1 union	0.259	4,180	0.40	0
$\mathcal{I}_{XFY}$	P2 family	0.079	4,322	0.38	0
$\mathcal{I}^{\star}_{XFY}$	P2/P3 (exact block)	0.000	0	1.00	0

6.4 Quantum-Workflow Coverage

All 165 cells are present and all nine prespecified acceptance checks pass. Clean cells trigger no sensor. Benign transpilation and the common-unitary rewrite change serialized provenance while preserving semantic kernels, algebraic properties and outputs in every cell: a raw hash difference needs policy-aware adjudication. Every data-dependent circuit mutation changes the semantic kernel; the mild RZ mutation changes no prediction, the stronger one changes predictions in 2/15 cells and the repetition change in 11/15. Asymmetric and diagonal-eroded matrices are detected algebraically in every cell and change no prediction; the PSD-preserving mixture is symmetric, unit-diagonal and PSD in all cells, so algebraic validation is exactly blind, while trusted provenance and semantic comparison expose all 15 substitutions and output monitoring reacts in 1/15. Both shot emulators produce non-zero repeated-estimation discrepancy in all cells and output changes in 7/15 and 3/15 (Fig. 3). The prototype's six envelopes verify and its eight acceptance checks pass: clean and approved transpilation are allowed, parameterized mutation and PSD-preserving substitution are blocked, prior-preserving label corruption is blocked at the evaluation contract despite a zero class-prior delta, and the approved 256-shot emulator is held.

6.5 Secondary Model-Impact Profile

The ZZ fidelity kernel has the largest mean conclusion impact in the frozen ID suite; its within-split difference to SVC is positive in 5/5 clusters at every dimension, positive in 7/8 environments at dimension 8 and 8/8 at 10 and 12, and correlated with clean headroom ($r = 0.714$, exploratory). Preregistered ablations (Fig. 1(b), supplement) show that the direction is stable but the magnitude is conditional: standardizing the quantum branch shrinks the difference to about a

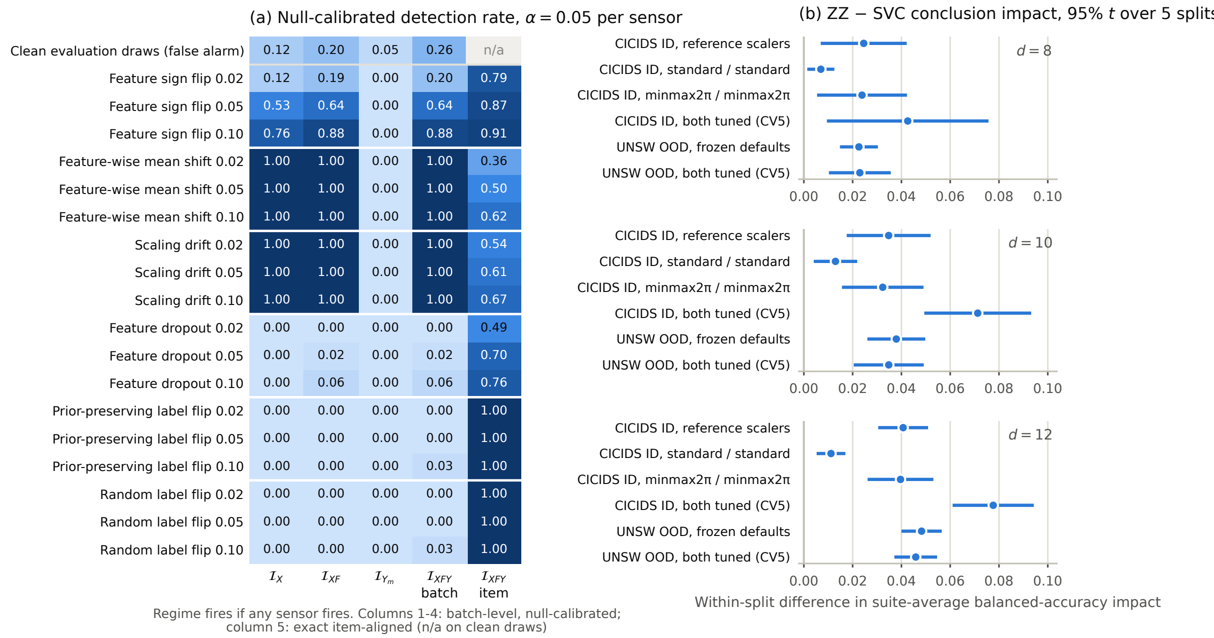

Fig. 1. Preregistered gates of artifact 1.1.0. (a) Null-calibrated detection rate of each information regime on the frozen interventions with the empirical false-alarm rate of disjoint clean evaluation draws in the first row; the first four columns are batch-level unions of per-sensor calibrated rules, the last column is exact item-aligned detection (undefined on clean draws). (b) Secondary within-split ZZ-minus-SVC conclusion-impact difference with 95% t intervals over five split clusters under three scaler configurations and under cross-validated tuning of both learners.

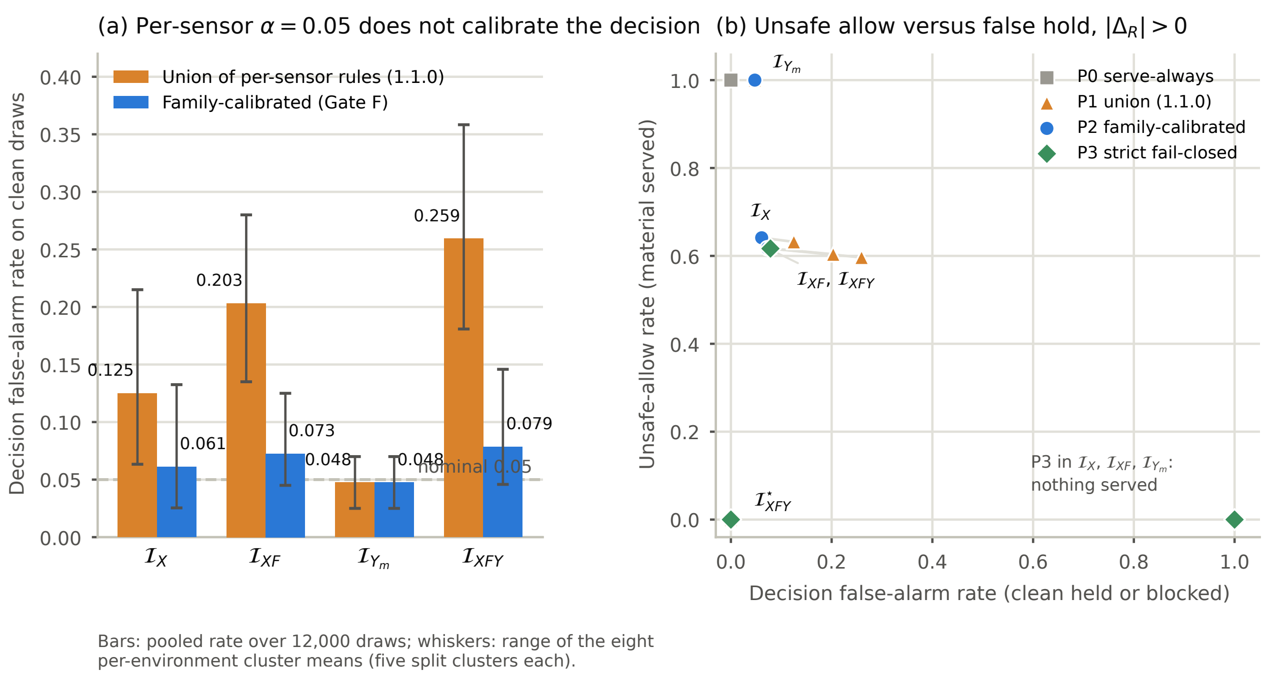

Fig. 2. Preregistered gates of artifact 1.2.0. (a) Decision-level false-alarm rate on 12,000 disjoint clean draws of the uncalibrated union rule and of the family-calibrated rule per regime; whiskers span the eight per-environment cluster means. (b) Unsafe-allow rate against decision false-alarm rate for every policy and regime at the primary materiality threshold; the trusted item-aligned regime is the only point at the origin, and the strict policy serves nothing where the label boundary is unverified.

third, matched min–max scaling leaves it unchanged, and cross-validated tuning of both learners widens it in CICIDS ID and leaves UNSW temporal OOD unchanged. Every configuration reproduces the exact label-path invariances. The profile is retained only to show that the primary result does not depend on the choice of model; it is not a quantum-advantage or vulnerability ordering.

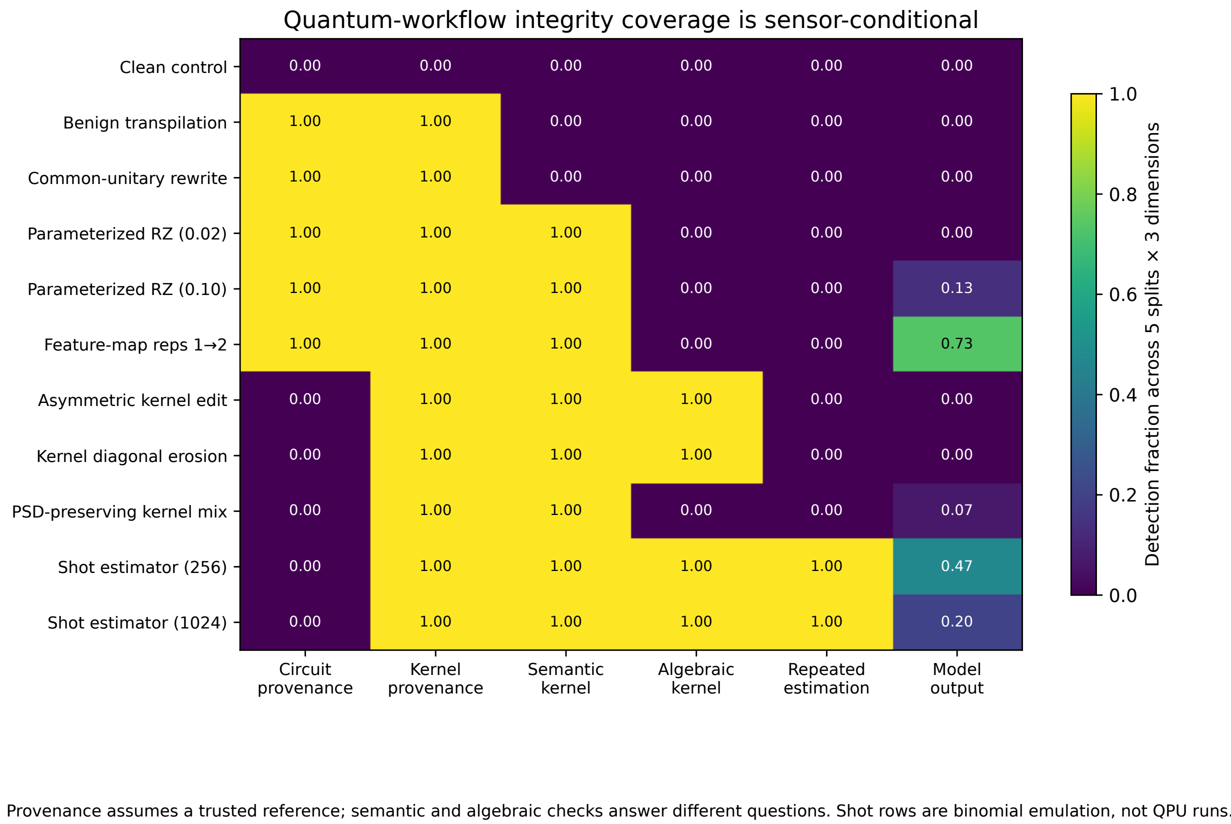

Fig. 3. Sensor response across clean and semantics-preserving controls, circuit mutations, kernel corruptions and finite-shot emulators over 165 simulator cells. Coverage is conditional on the evidence exposed to each sensor; output invariance does not imply workflow integrity.

7 Integrity-Audit Contract

The evidence yields a reporting and enforcement contract: every integrity claim names the boundary and protected asset, the adversary or failure class and what remains fixed, the information regime and the trusted references it assumes, the sensor, threshold and decision-level false-alarm budget, the covered interventions and the structural and calibrated residual blind regions, and the response. For the label path the minimum bundle is item-level label provenance plus a joint label-prediction audit; for the studied quantum boundary it is authenticated circuit/parameter provenance, an approved-transpilation policy, semantic probe kernels, algebraic checks, repeated-estimation evidence and fail-closed handling of missing evidence. `allow` requires every mandatory contract to pass and, under the strict policy, every protected boundary to be covered; unresolved stochastic deviation or statistical evidence yields `hold`; an exact violation yields `block`. Missing evidence is never converted into assurance.

8 Discussion and Limitations

8.1 Security Interpretation

A metric can move while predictor output is bit-for-bit unchanged, and it can move upward. Calling either model degradation assigns failure to the wrong component and suggests the wrong defence: input robustness cannot repair corrupted ground truth; provenance and joint validation can. Zero sensor response is informative only if the sensor receives evidence capable of distinguishing the states, and non-zero separability is informative only if the auditor holds a null tight enough to see it. The end-to-end result makes the consequence concrete: with the evidence a batch-level auditor normally has, three fifths of the materially changed results in this suite would be served under a calibrated policy, and the share is set by the information regime, not by the threshold. The lever is provenance.

8.2 Validity Boundaries

The study has the following limits. Three datasets and eight fixed scale/temporal settings are not sampled deployments; five split clusters measure within-environment resampling only; most gates use 128/128 samples; balanced numeric-only staging does not reproduce operational prevalence. Both quantum evaluators are ideal statevector simulators with binomial shot emulation. SVC and QSVC keep pipeline-specific scalers, so their comparison does not isolate kernel geometry. Exact blindness is a capability result under constructed perturbations and estimates no prevalence. The calibrated rates are fixed-design, simulator-only sensitivities: draws overlap within a pool half, the calibration and evaluation halves are different rows of one table, the family rule's residual excess over nominal is real and reported, the largest environment reuses a small pool, and the KS sensitivity rests on cluster structure that an adversary preserving it would evade. The trusted item-aligned regime is a trust assumption, not a stronger detector: if the reference can be rewritten with the asset, Proposition 6 applies. The prototype is a local process without authenticated provider records, persistence or incident operations, and its hash chain gives neither identity nor non-repudiation. Context-conditioned runtime calibration under non-stationarity, abstention with recovery, real-QPU execution and the service-level evaluation of enforcement are the subject of a separate study.

9 Conclusion

Hybrid workflow integrity cannot be reduced to a robustness number. An intervention can change a reported conclusion without changing the model, a sensor can stay silent because its view is identical, and a separable change can stay undetected because the auditor holds no anchor. We formalized this as observational indistinguishability under explicit information sets and trusted references, derived monotone and exactly closable blind regions with the label path as a protected boundary, reproduced them across three datasets and eight environments under null calibration, calibrated the decision rather than the sensors, and measured end to end how many materially changed results each regime would serve. The requirement is precise: every integrity claim must state its boundary, view, trusted roots, decision-level budget, residual blind regions and response, and assurance is granted only when the evidence the claim requires is present.

Data, Code, and Ethics Statement

The review artifact contains source code, a locked environment, tests, compact derived tables, figures, six SHA-256 evidence manifests and exact reproduction commands, including the preregistrations of the 1.1.0 and 1.2.0 gates. Raw CICIDS2017, UNSW-NB15 and ToN-IoT data are not redistributed; their public sources, expected hashes and deterministic staging are documented. Version 1.2.0 is archived at Zenodo under the concept DOI 10.5281/zenodo.22550852 (version DOI in the artifact metadata); source code is released under Apache-2.0 and derived evidence, figures and documentation under CC BY 4.0. The study uses previously released network-security benchmarks and involves no new human-participant, animal or personal-data collection.

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